

1. Objective:

To build and evaluate a fruit classification system using Weka by comparing five different machine learning algorithms on a synthetic dataset with realistic, overlapping features.

2. Methodology:

- Dataset: An ARFF dataset of 578 instances representing three classes: Apples, Bananas, and Oranges.
- Features: Four numeric attributes: weight (g), roundness (1-10), redness (1-10), and yellowness (1-10). The data included intentional noise, such as green apples and small oranges, to test model robustness.
- Evaluation Method: 10-Fold Cross-Validation was used to ensure unbiased testing and prevent overfitting.

3. Results Summary:

The classifiers were ranked by their overall accuracy in correctly identifying the fruits:

Classifier	Correctly Classified	Accuracy (%)	Precision (Weighted)	Recall (Weighted)	F-Measure (Weighted)
ClassificationViaRegression	578 / 578	100%	1	1	1
MultiClassClassifier (Logistic)	574 / 578	99.31%	0.993	0.993	0.993
SimpleLogistic	573 / 578	99.13%	0.991	0.991	0.991
MultilayerPerceptron (Neural Net)	~575 / 578	99.50%	0.995	0.995	0.995
NaiveBayes	556 / 578	96.19%	0.962	0.962	0.962

4. Key Findings & Analysis:

- Most Effective Models: ClassificationViaRegression and the logistic/neural models performed best with 99-100% accuracy. They successfully weighed correlated features to create complex boundaries between the overlapping data points for Apples and Oranges.
- Least Effective Model: NaiveBayes scored the lowest at 96.19%. Because it assumes all features are independent, it struggled to separate overlapping items such as heavy oranges and round apples.
- Confusion Matrix Insights: Apple vs. Orange Overlap: All 22 misclassifications in the Naive Bayes model occurred between Apples and Oranges, with 7 Apples labeled as Oranges and 15 Oranges labeled as Apples.
 - Banana Accuracy: Bananas were classified with 100% accuracy across almost all models. Their distinct shape, low roundness, and high yellowness made them easy to distinguish from round fruits.

5. Conclusion:

Regression and logistic models are very effective at handling noisy, correlated real-world data. While simpler probabilistic models like NaïveBayes provide a good baseline, they are prone to errors when class features overlap significantly.